Computer Science - K-12 Level Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. In Python 3, what is the output of ['Hi!'] * 4?

'Hi!', 'Hi!', 'Hi!', 'Hi!'

'Hi!' * 4

   (a) Error
   (b) None of the above.

2. A large Java program was tested extensively, and no errors were found. What can be concluded?
   (a) All of the preconditions in the program are correct.
   (b) All of the postconditions in the program are correct.
   (c) The program may have bugs.
   (d) Every method in the program may safely be used in other programs.

3. Let l = [1,2,2,3,4]. In Python3, what is a possible output of set(l)?
   (a) {1}
   (b) {1,2,2,3,4}
   (c) {1,2,3,4}
   (d) {4,3,2,2,1}

4. In Python 3, what is the output of print list[1:3] if list = ['abcd', 786 , 2.23, 'john', 70.2]?

786 , 2.23, 'john', 70.2

   (a) abcd

786, 2.23

   (b) None of the above.

5. What is the output of the statement "a" + "ab" in Python 3?
   (a) Error
   (b) aab
   (c) ab
   (d) a ab

True / False (10 marks)

Answer True or False

6. True or False: In Python, accessing the first element of a tuple always results in an error.

7. True or False: The output of the expression 4*1**4 in Python is 16.

8. True or False: The boolean expression a[i] == max || !(max != a[i]) simplifies to a[i] != max.

9. True or False: In Python 3, the function float(x) is used to convert a string to an integer.

10. True or False: A denial-of-service (DoS) attack requires a larger number of computers than a distributed denial-of-service (DDoS) attack.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Explain why adding the decimal numbers 5 and 3 in a computer program using only 3 bits results in 0. Discuss the concept of overflow and its implications in binary arithmetic.

12. Consider the Python list b = [11, 13, 15, 17, 19, 21]. Explain how the slicing operation b[::2] works and discuss the specific elements it retrieves from the list.

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